Eli Ben-Michael

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EDUCATION

University of California, Berkeley, Berkeley, CA — December 2020
PhD in Statistics, Dissertation Committee: Avi Feller, Peng Ding, Jesse Rothstein, Bin Yu

Columbia University, Columbia College, New York, NY — May 2016
Bachelor of Arts, *Summa Cum Laude*, Computer Science and Statistics

ACADEMIC APPOINTMENTS

Carnegie Mellon University, Pittsburgh, PA — August 2022 -
Heinz College of Information Systems & Public Policy
Department of Statistics & Data Science
Heinz College Dean's Career Development Assistant Professor — August 2026 -
Assistant Professor — August 2022 - August 2026

Harvard University, Cambridge, MA — January 2021 - August 2022
Department of Statistics
Institute for Quantitative Social Science
Postdoctoral Fellow

FORTHCOMING AND PUBLISHED ARTICLES

Schwartz, A. L., T. Bulat, **E. Ben-Michael**, E. Wang, J. Zhu, E. Boudreau, S. Chhatre, S. Kim, D. Pagnotti, S. Riaz, A. Sutherland, and A. Navathe (2026). Service-Level Utilization in Risk-Based Contracts and a Benchmarking Approach for Fee-Schedule Reforms. *JAMA Health Forum 7*(10)

Shen, A. A., **E. Ben-Michael**, A. Feller, L. Keele, and J. Murray (2026). Forest kernel balancing weights: Outcome-guided features for causal inference. *Statistics in Medicine 45*(20-22), e70720

Sundberg, J., R. Ghani, **E. Ben-Michael**, and E. Kennedy (2026). Learning Who to Treat When Treatment is Missing. In *Conference on Uncertainty in Artificial Intelligence*. arXiv:2607.14346

Arbour, D., **E. Ben-Michael**, A. Feller, A. Lal, and L.-H. Yuan (2026). AI-Assisted Variance Reduction in Randomized Experiments. In *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. arXiv:2606.08853

Keele, L., **E. Ben-Michael**, M. Lenard, and L. Page (2026). Balancing weights for estimating treatment effects in educational studies. *Journal of Research on Educational Effectiveness 19*(2), 375–402

Jiang, Z., **E. Ben-Michael**, D. J. Greiner, R. Halen, and K. Imai (2026). Longitudinal causal inference with selective eligibility: Application to pretrial risk assessment. *The Annals of Applied Statistics 20*(3), 1917–1941

Lin, V., L.-P. Morency, and **E. Ben-Michael** (2026). Omitted Variable Bias in Language Models Under Distribution Shift. In *International Conference on Machine Learning*. arXiv:2602.16784

Gemmill, A., A. Franks, A. Feller, E. A. Stuart, **E. Ben-Michael**, and S. O. Bell (2026). Methodological considerations for investigating the impact of abortion restrictions on outcomes using aggregate panel data. *American Journal of Epidemiology 195*(7), 2094–2099

Huang, M., **E. Ben-Michael**, M. McHugh, and L. Keele (2026). Sensitivity analysis for clustered observational studies with an application to the effectiveness of magnet nursing hospitals. *Journal of the Royal Statistical Society Series A: Statistics in Society*, qnag002

Jia, Z., **E. Ben-Michael**, and K. Imai (2026). Bayesian Safe Policy Learning with Chance Constrained Optimization: Application to Military Security Assessment during the Vietnam War. *Journal of the*

Royal Statistical Society, Series A: Statistics in Society 189, 1448–1472

Ben-Michael, E., D. J. Greiner, M. Huang, K. Imai, Z. Jiang, and S. Shin (2025). Does AI help humans make better decisions? A statistical evaluation framework for experimental and observational studies. *Proceedings of the National Academy of Sciences 122*(38)

Lin, V., L. P. Morency, and **E. Ben-Michael** (2025). Isolated Causal Effects of Natural Language. In *International Conference on Machine Learning*

Ben-Michael, E., D. J. Greiner, K. Imai, and Z. Jiang (2025). Safe Policy Learning through Extrapolation: Application to Pre-trial Risk Assessment. *Journal of the American Statistical Association 120*(551), 1386–1399

Sun, L., **E. Ben-Michael**, and A. Feller (2025). Using Multiple Outcomes to Improve the Synthetic Control Method. *Review of Economics and Statistics*

Gemmill, A., A. M. Franks, S. Anjur-Dietrich, A. Ozinsky, D. Arbour, E. A. Stuart, **E. Ben-Michael**, A. Feller, and S. O. Bell (2025). US Abortion Bans and Infant Mortality. *JAMA 333*(15), 1315–1323
American Statistical Association 2025 Outstanding Statistical Application Award winner

Bell, S. O., A. M. Franks, D. Arbour, S. Anjur-Dietrich, E. A. Stuart, **E. Ben-Michael**, A. Feller, and A. Gemmill (2025). US Abortion Bans and Fertility. *JAMA 333*(15), 1324–1332
American Statistical Association 2025 Outstanding Statistical Application Award winner

Ben-Michael, E., A. Feller, R. Kelz, and L. Keele (2025). Estimating racial disparities in emergency general surgery. *Journal of the Royal Statistical Society Series A: Statistics in Society 188*(4), 1125–1148

Lin, V., **E. Ben-Michael**, and L. P. Morency (2024). Optimizing Language Models for Human Preferences is a Causal Inference Problem. In *Conference on Uncertainty in Artificial Intelligence*

Sun, L., **E. Ben-Michael**, and A. Feller (2024). Temporal Aggregation for the Synthetic Control Method. In *AEA Papers and Proceedings*, Volume 114, pp. 614–617

Ben-Michael, E., L. Page, and L. Keele (2024). Approximate balancing weights for clustered observational study designs. *Statistics in Medicine 43*(12), 2332–2358

Ben-Michael, E., K. Imai, and Z. Jiang (2024). Policy Learning with Asymmetric Counterfactual Utilities. *Journal of the American Statistical Association 119*(548), 3045–3058

Ben-Michael, E., A. Feller, and E. Hartman (2024). Multilevel Calibration Weighting for Survey Data. *Political Analysis 32*(1), 65–83

Lin, V., L. P. Morency, and **E. Ben-Michael** (2023). Text-Transport: Toward Learning Causal Effects of Natural Language. In *Conference on Empirical Methods in Natural Language Processing*

Ben-Michael, E. and L. Keele (2023). Using Balancing Weights to Target the Treatment Effect on the Treated when Overlap is Poor. *Epidemiology 4*(34), 637–644

Soriano, D., **E. Ben-Michael**, P. J. Bickel, A. Feller, and S. D. Pimentel (2023). Interpretable Sensitivity Analysis for Balancing Weights. *Journal of the Royal Statistical Society, Series A: Statistics in Society*

Ben-Michael, E., A. Feller, and J. Rothstein (2023). Varying Impacts of Letters of Recommendation on College Admissions. *The Annals of Applied Statistics 17*(4), 2843–2864
American Statistical Association Social Statistics Section Student Paper Award winner

Lu, B., **E. Ben-Michael**, A. Feller, and L. Miratrix (2023). Is it who you are or where you are? Accounting for compositional differences in cross-site treatment variation. *Journal of Educational and Behavioral Statistics 48*(4), 420–453
American Statistical Association Editors Choice 2023

Ben-Michael, E., D. Arbour, A. Feller, A. Franks, and S. Raphael (2023). Estimating the effects of a California gun control program with Multitask Gaussian Processes. *The Annals of Applied Statistics 17*(2), 985–1016

Keele, L. J., **E. Ben-Michael**, A. Feller, R. Kelz, and L. Miratrix (2023). Hospital Quality Risk Standardizations via Approximate Balancing Weights. *The Annals of Applied Statistics 17*(2), 901–

Ben-Michael, E., A. Feller, and J. Rothstein (2022). Synthetic controls with staggered adoption. *Journal of the Royal Statistical Society. Series B: Statistical Methodology 84*(2), 351–381
SF ASA Student Travel Award winner, Thomas R. Ten Have Poster Award runner up

Haber, N. A., E. Clarke-Deelder, A. Feller, E. R. Smith, J. A. Salomon, B. MacCormack-Gelles, E. M. Stone, C. Bolster-Foucault, J. R. Daw, L. A. Hatfield, C. E. Fry, C. B. Boyer, **E. Ben-Michael**, C. M. Joyce, B. S. Linas, I. Schmid, E. H. Au, S. E. Wieten, B. Jarrett, C. Axfors, V. T. Nguyen, B. A. Griffin, A. Bilinski, and E. A. Stuart (2022). Problems with evidence assessment in COVID-19 health policy impact evaluation: A systematic review of study design and evidence strength. *BMJ Open 12*(1)

Ben-Michael, E., A. Feller, and J. Rothstein (2021). The Augmented Synthetic Control Method. *Journal of the American Statistical Association 116*(536), 1789–1803

Ben-Michael, E., A. Feller, and E. Stuart (2021). A trial emulation approach for policy evaluations with group-level longitudinal data. *Epidemiology 32*, 533–540

Elser, H., **E. Ben-Michael**, D. Rehkopf, S. Modrek, E. A. Eisen, and M. R. Cullen (2019). Layoffs and the mental health and safety of remaining workers: a difference-in-differences analysis of the US aluminium industry. *Journal of Epidemiology and Community Health 73*, 1094–1100

PREPRINTS AND WORKING PAPERS

Ben-Michael, E. and Z. Branson (2026). AI-assisted design and analysis of experiments with unstructured treatments. arXiv:2610.33818

Ben-Michael, E., A. Feller, and L. Keele (2026). A Weighting Framework for Clusters as Confounders in Observational Studies. arXiv:2602.05041

Raji, I. D., L. T. Liu, A. Zhou, L. Guerdan, J. Hullman, D. Malinsky, B. Wilder, S. Zhang, H. Adam, A. Coston, B. Laufer, E. Nwankwo, M. Zanger-Tishler, **E. Ben-Michael**, A. Feller, T. Gillis, S. Guha, D. Ho, L. Hu, K. Imai, S. Kapoor, J. Loftus, R. Nabi, J. C. Perdomo, M. Salganik, M. Sendak, B. Ustun, S. Venkatasubramanian, A. Wang, and A. Wilson (2026). Bridging Predictions and Interventions: An Integrated Framework for Automated Decision-Systems. arXiv:2606.25668

Ben-Michael, E. (2025). Partial identification via conditional linear programs: estimation and policy learning. arXiv:2506.12215

Liu, L. T., I. D. Raji, A. Zhou, L. Guerdan, J. Hullman, D. Malinsky, B. Wilder, S. Zhang, H. Adam, A. Coston, B. Laufer, E. Nwankwo, M. Zanger-Tishler, **E. Ben-Michael**, S. Barocas, A. Feller, M. Gerchick, T. Gillis, S. Guha, D. Ho, L. Hu, K. Imai, S. Kapoor, J. Loftus, R. Nabi, A. Narayanan, B. Recht, J. C. Perdomo, M. Salganik, M. Sendak, A. Tolbert, B. Ustun, S. Venkatasubramanian, A. Wang, and A. Wilson (2025). Bridging Prediction and Intervention Problems in Social Systems. arXiv:2507.05216

Levis, A. W., **E. Ben-Michael**, and E. H. Kennedy (2024). Intervention effects based on potential benefit. arXiv:2405.08727

Zhang, Y., **E. Ben-Michael**, and K. Imai (2022). Safe Policy Learning under Regression Discontinuity Designs with Multiple Cutoffs. arXiv:2208.13323
Yi Zhang: American Statistical Association Social Statistics Section Student Paper Award winner

Ben-Michael, E., D. A. Hirshberg, A. Feller, and J. Zubizarreta (2021). The Balancing Act in Causal Inference. arXiv:2110.14831

DISCUSSIONS AND COMMENTS

Ben-Michael, E. and A. Feller (2026). Discussion of "Matrix Completion When Missing Is Not at Random and Its Applications in Causal Panel Data Models". Invited discussion at JSM 2025 arXiv:2602.21314

BOOK CHAPTERS

Ben-Michael, E., M. L. Doucette, A. Feller, A. D. McCourt, and E. Stuart (2026). Statistical methods to estimate the impact of gun policy on gun violence. In C. Loeffler, J. L. Rosenberger, and L. Xue (Eds.), *Gun Violence: The Statistical Issues* (1 ed.). Chapman and Hall/CRC

Cohn, E. R., **E. Ben-Michael**, A. Feller, and J. R. Zubizarreta (2023). Balancing Weights for Causal Inference. In J. R. Zubizarreta, E. A. Stuart, D. S. Small, and P. R. Rosenbaum (Eds.), *Handbook of Matching and Weighting Adjustments for Causal Inference* (1 ed.)., Chapter 16, pp. 293–312. Boca Raton, FL: Chapman & Hall/CRC

OPEN SOURCE STATISTICAL SOFTWARE

`augsynth`: `R` implementation of the augmented synthetic control method

`multical`: `R` package for multilevel calibration weighting

`balancer`: `R` package for balancing weights in observational studies

AWARDS AND HONORS

American Statistical Association Outstanding Statistical Application Award 2025

American Statistical Association Editors Choice, Journal of Educational and Behavioral Statistics 2023

American Causal Inference Conference Travel Award 2022

American Statistical Association Social Statistics Section Student Paper Award 2021

American Statistical Association, San Francisco Bay Area Chapter, Student Travel Award 2020

Department fellowship, Department of Statistics, U.C. Berkeley 2018
Two years of funding through RTG grant: Advancing Machine Learning - Causality and Interpretability

Phi Beta Kappa, Columbia University 2016

Computer Science Department Award, Columbia University 2016

RESEARCH FUNDING

Statistical Innovations for Clustered Observational Studies 2021-2025
Department of Education, Institute of Education Sciences R305D210014 $899,023
Principal Investigators: Luke Keele, Lindsay Page
Co-Investigator

Statistical Innovations for Balancing Weight Methods in Education Research 2024-2027
Department of Education, Institute of Education Sciences R305D240036 $899,779
Principal Investigators: Luke Keele, Avi Feller, Eli Ben-Michael
Co-Principal Investigator

TEACHING

Carnegie Mellon University

36-309/36-749 Experimental Design for Behavioral & Social Sciences	Fall 2024-2026
95-796: Statistics for IT Managers	Fall 2024-2026
90-777: Intermediate/Accelerated Statistics	Fall 2023-2025
90-739: Systems Synthesis	Spring 2024
36-490/36-497 Statistics Research (and Corporate) Capstones	Fall 2023
36-402: Advanced Data Analysis	Spring 2023

U.C. Berkeley *Graduate Student Instructor*

Stat 232: Experimental Design with Sam Pimentel	Fall 2018
Stat 159/259: Reproducible and Collaborative Data Science with Fernando Perez	Fall 2017

SERVICE

ASA Social Statistics Section Program Chair (Chair-Elect in 2025-2026)	2026-2027
2027 American Causal Inference Conference Program Committee	2026-2027
2027 International Conference on Health Policy Statistics Scientific Organizing Committee	2026-2027
Statistics Faculty Search Committee, CMU	2025-2026
Statistics PhD admissions Committee, CMU	2024-2025
ICHPS 2025 Student Paper Award Committee	2024
Organizer, Heinz Informal Lunch Seminar	2023 -
Joint Statistics and Public Policy PhD admissions committee, CMU	2022 -
PhD admissions committee, Department of Statistics, UC Berkeley	2020
Co-president of the Berkeley Statistics Graduate Student Association	2018 - 2019

Reviewer for *American Economic Review: Insights*, *American Journal of Epidemiology*, *Annals of Applied Statistics*, *Biometrical Journal*, *Biometrics*, *Biometrika*, *Econometrica*, *Epidemiology*, *INFORMS Journal on Data Science*, *Journal of the American Statistical Association*, *Journal of Applied Econometrics*, *Journal of Business & Economic Statistics*, *Journal of Causal Inference*, *Journal of Econometrics*, *Journal of Educational and Behavioral Statistics*, *Journal of Quantitative Criminology*, *Journal of the Royal Statistical Society, Series A, B, and C*, *Management Science*, *Operations Research*, *Political Analysis*, *Population Health Metrics*, *Quantitative Economics*, *Review of Economics and Statistics*, *Statistics in Medicine*, and *Statistics and Public Policy*

PRESENTATIONS

2026

Invited: Simons Institute Workshop on Bridging Prediction and Intervention Problems in Social Systems; Stanford Political Science Methods Workshop; NYU Political Science Methods Workshop; Columbia Political Science Methods Workshop; Northwestern HCI+D Symposium on Validating Generative AI-Based Social Sciences; Columbia Data Science Institute Frontiers: Machine-Generated Experimental Designs and The Future of Social Science; Isaac Newton Institute Workshop on Causality and Machine Learning; International Chinese Statistical Association – Canada Chapter Symposium; University of Michigan Causal AI for Decision Making Workshop; INFORMS 2027

Contributed: American Causal Inference Conference

2025

Invited: International Conference on Health Policy Statistics; University of Washington Center for Statistics and the Social Sciences; University of Washington Department of Statistics; Pennsylvania State University Center for Social Data Analytics; Joint Statistical Meetings; UC Irvine Statistics Seminar

Contributed: American Causal Inference Conference; Annual Meeting of the Society for Political Methodology

2024

Invited: Boston University Department of Economics; Columbia University Department of Political Science; Banff International Research Station: Bridging Prediction and Intervention

Problems in Social Systems; Johns Hopkins University Department of Biostatistics

Contributed: Annual Meeting of the Society for Political Methodology; Joint Statistical Meetings

2023

Invited: International Conference on Health Policy Statistics; Cowles Foundation Conference on Econometrics; Joint Statistical Meetings; University of Pittsburgh Department of Epidemiology

Contributed: American Causal Inference Conference

2022

Invited: London School of Hygiene and Tropical Medicine, Centre for Statistical Methodology; Joint Statistical Meetings; Linkedin; National Research Conference on Firearm Injury Prevention; IMS International Conference on Statistics and Data Science; International Conference on Computational and Methodological Statistics; Bank of England

Contributed: American Causal Inference Conference

2021

Invited: Johns Hopkins CICADAS Seminar; RAND Center for Causal Inference Symposium; Johns Hopkins Causal Inference Seminar; Joint Statistical Meetings (*ASA Social Statistics Section Student Paper Award Winner*); Harvard Applied Statistics Workshop; Berkeley Machine Learning and Science Forum

Contributed: Asian Political Methodology Meetings; ENAR Spring Meeting; Annual Meeting of the Society for Political Methodology

2020

Invited: Online Causal Inference Seminar

Contributed: Annual Meeting of the Society for Political Methodology; Joint Statistical Meetings (*SFASA Student Travel Award winner*); Econometric Society World Congress; Association for Public Policy Analysis & Management Fall Research Conference; BigSurv20

2019

Contributed: Society for Research on Educational Effectiveness; Atlantic Causal Inference Conference (Thomas R. Ten Have Poster Award runner up); Berkeley-Stanford Econometrics Jamboree

2018

Contributed: 7th Causal Inference Workshop at UAI; European Winter Meeting of the Econometric Society

OTHER EXPERIENCE

Uber, New York, NY — Summer 2019
Data Science Intern

Walmart Labs, Sunnyvale, CA — Summer 2017
Machine Learning Scientist Intern

Knewton, New York, NY — Summer 2016
Data Science Intern